Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/05/2018
PI Name:	Mehta, Satish Ph.D.		
Project Title:	Determination of Roles of Microgravity and Ionizing Radiation on the Reactivation of Epstein-Barr Virus		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-5459
Organization Name:	Enterprise Advisory Services, Inc.		
PI Address 1:	NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058-2720	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	10/01/2013	**End Date:**	10/01/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:			
Flight Assignment:	NOTE: Extended to 9/30/2019 (using 10/01/2019 as end date for reporting purposes) per F. Hernandez/ARC (Ed., 7/10/2018) NOTE: Extended to 9/30/2018 (using 10/01/2018 as end date) per F. Hernandez/ARC (Ed., 9/20/17) NOTE: Extended to 9/30/2017 per F. Hernandez/ARC (Ed., 9/28/16)		
Key Personnel Changes/Previous PI:	July 2018: no changes		
COI Name (Institution):	Bloom, David Ph.D. (University of Florida) Pierson, Duane Ph.D. (NASA Johnson Space Center) Stowe, Raymond Ph.D. (Microgen, LLC) Wu, Honglu Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We will study the effects of simulated microgravity and ionizing radiation on the complex process of virus reactivation that is intimately associated with the way cells respond to external stress. Tools to evaluate the effects of radiation and simulated microgravity will include: both viral and cellular gene expression, viral infectivity and viral titer, and DNA methylation. In addition, this proposal aims to increase our understanding of mechanisms of Epstein-Barr Virus (EBV) reactivation and therefore, has the potential to discover new therapeutic targets for diseases caused in the general population by EBV. Therefore this proposal appropriately addresses: 1. Ground-based Research Designed to Lead to Space Flight; 2. Cell Biology and Microbiology: This section of the NASA Research Announcement (NRA) solicits NOVEL ground-based experiments that use cell cultures, tissue cultures, microbiological organisms, or small invertebrate organisms to elucidate the basic mechanisms living systems use to respond to changes in gravity, i.e., how are changes sensed and transduced to alter gene expression and the structure and function of the musculoskeletal system. The reactivation of all herpes viruses is known to involve 2 components or stages: the initial initiating molecular events that cause reactivation of the latent genomes to progress to the production of reactivating virus, and then the propagation of these reactivating viruses to produce clinical disease. The second part of this process undoubtedly involves overcoming components of the acquired immune response if clinical disease is to result. In order to understand reactivation from a mechanistic point of view, these two processes need to be examined independently. In the proposed studies, we will be focusing only on the initial events that microgravity and ionizing radiation initiate in latent cells to induce the first stage of (molecular) reactivation. EBV reactivation may not be completely homologous to Varicella zoster virus (VZV) and Cytomegalovirus (CMV) reactivation. The important point here though is that by focusing on EBV, which has a well-established cell culture system of latency, we can study the mechanistic effects of microgravity and ionizing radiation in inducing EBV reactivation. This alone will provide critical and important information regarding the effects of these inducers on EBV reactivation in humans. Clearly, after these molecular events are established in EBV, possible parallel mechanisms in VZV and CMV can be investigated in future studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal aims to increase our understanding of mechanisms of Epstein-Barr Virus (EBV) reactivation and therefore has the potential to discover new therapeutic targets for diseases caused in the general population by EBV.
Task Progress:	In general, our work shows that gamma rays are more effective than protons, carbon ions, and iron ions in inducing latent virus reactivation, though these high energy particles did induce more sustained and later reactivation of EBV lytic gene transcription. An oral presentation was made at the annual American Society for Gravitational and Space Research (ASGSR) meeting in Seattle in October 2017—see Bibliography section. A manuscript submitted to International Journal of Molecular Sciences is under review: “Reactivation of Latent Epstein-Barr Virus: A Comparison after Exposure to Gamma, Proton, Carbon, and Iron Radiation”—see also Bibliography section.
Bibliography Type:	Description: (Last Updated: 04/15/2025)
Abstracts for Journals and Proceedings	Mehta SK, Bloom DC, Plante I, Stowe R, Feiveson AH, Renner A, Dhummakupt A, Markan M, Zhang Y, Wu H, Scoles B, Cohen JI, Crucian B, Pierson DL. "Reactivation of latent Epstein-Barr virus; a comparison after gamma rays and proton treatment." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Articles in Peer-reviewed Journals	Mehta SK, Bloom DC, Plante I, Stowe R, Feiveson AH, Renner A, Dhummakupt A, Markan M, Zhang Y, Wu H, Scoles B, Cohen JI, Crucian B, Pierson DL. "Reactivation of latent Epstein-Barr Virus: A comparison after exposure to gamma, proton, carbon, and iron radiation." International Journal of Molecular Sciences. 2018 Sep 28;19(10):2961. https://doi.org/10.3390/ijms19102961 ; PubMed PMID: 30274169; PubMed Central PMCID: PMC6213004 , Sep-2018